



2019 HSC Course Assessment Task 3 (Trial Examination)

27th of June, 2019

- Working time – 3 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

- Mark your answers on the answer sheet provided (numbered as page 13)

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

Class (please ✓)

○ 12M4C – Miss Lee

[illegible]

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer sheet provided.

Questions	Marks
1. Write $\frac{10i}{1+3i}$ in the form $a+ib$, where a and b are real. (A) $1-3i$ (B) $1+3i$ (C) $3+i$ (D) $3-i$	1
2. The equation $x^3+px+q=0$ has a double root 1. What is the value of $p+q$? (A) 1 (B) -1 (C) 2 (D) -3	1
3. The value of $i+i^2+i^3+\dots+i^{2019}$ (A) i (B) 1 (C) $-i$ (D) -1	1
4. The value of $\lim_{h \rightarrow 0} \frac{f(a+3h)-f(a)}{h}$ is (A) $f'(3a)$ (B) $3f'(3a)$ (C) $3f'(a)$ (D) $\frac{f'(a)}{3}$	1
5. The eccentricity of the ellipse $\frac{x^2}{3^2+4^2}+\frac{y^2}{3^2}=1$ is (A) $\frac{3}{4}$ (B) $\frac{4}{9}$ (C) $\frac{4}{5}$ (D) $\frac{16}{25}$	1
6. $P(z)$ is a polynomial of degree 4. Which of the following statements must be false? (A) $P(z)$ has 4 real roots. (B) $P(z)$ has 2 real and 2 non real roots (C) $P(z)$ has 1 real and 3 non real roots (D) $P(z)$ has no real roots	1

7. Given the curve $y = f(x)$, then the curve $y = f(|x|)$ is represented by 1
- (A) A reflection of $y = f(x)$ in the y axis.
- (B) A reflection of $y = f(x)$ in the x axis.
- (C) A reflection of $y = f(x)$ in the y axis for $x \geq 0$.
- (D) A reflection of $y = f(x)$ in the x axis for $y \geq 0$.
8. How many vertical tangents can be drawn on the graph of $x^2 + y^2 + 4xy - 4 = 0$. 1
- (A) 1 (B) 2 (C) more than 2 (D) 0
9. Which of the following $\int \frac{dx}{\sqrt{7-6x-x^2}}$? 1
- (A) $\sin^{-1}\left(\frac{x-3}{2}\right) + c$
- (B) $\sin^{-1}\left(\frac{x+3}{2}\right) + c$
- (C) $\sin^{-1}\left(\frac{x-3}{4}\right) + c$
- (D) $\sin^{-1}\left(\frac{x+3}{4}\right) + c$
10. Given $F(x) = \int_a^x (x-t) \cos 3t dt$, then $F''(x)$ is 1
- (A) $(1-x) \cos 3x$
- (B) $\sin 3x$
- (C) $(1-3x) \cos 3x$
- (D) $\cos 3x$

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Commence a NEW page.	Marks
(a) Find $\int x^4 e^{x^5+3} dx$		1
(b) Use the substitution $u = \frac{1}{1+x^2}$ to evaluate $\int \frac{dx}{x(1+x^2)^2}$.		2
(c) Find $\int \frac{3x}{5x^2 - 4x + 2} dx$		3
(d) Given $I = \int x \sin^2 x dx$ and $J = \int x \cos^2 x dx$		
i. Show that $I + J = \frac{x^2}{2} + c_1$		1
ii. Find $J - I$		3
iii. Hence, or otherwise find I and J .		1
(e)		
i. Prove that if f is continuous function, then		2
$\int_0^a f(x) dx = \int_0^a f(a-x) dx.$		
ii. Hence, or otherwise show that		2
$\int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{4}.$		

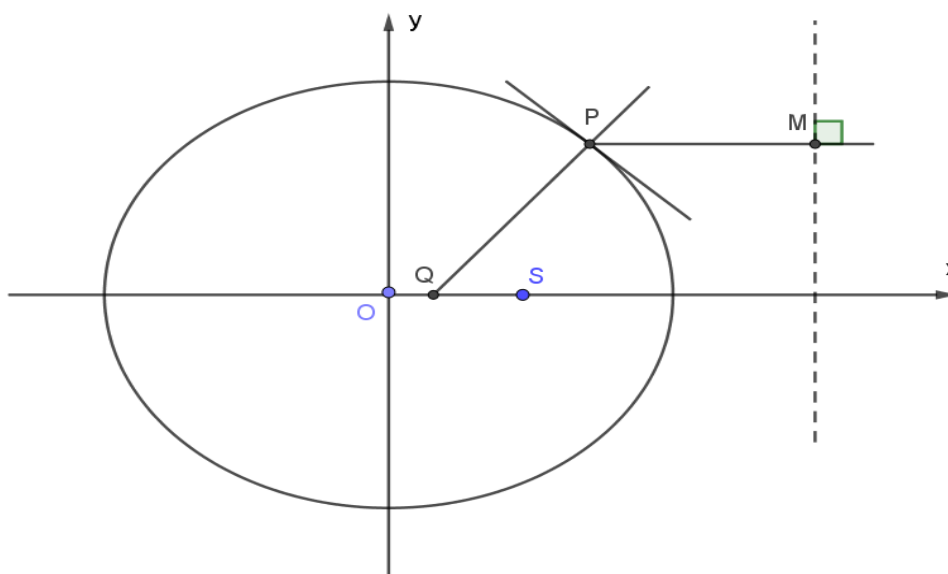
End of Question 11

Question 12 (15 Marks)

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Marks

- (a) Given $z = 1 - i$, find the values of w such that $w^2 = i + 3\bar{z}$ **2**
- (b) Find the Cartesian equation of the locus of a point P which represents the complex number z where $|z - 2i| = |z|$ **2**
- (c) z is a point in the first quadrant of the Argand diagram which lies on the circle $|z - 3| = 3$. Given $\arg(z) = \theta$, find $\arg(z^2 - 9z + 18)$ in terms of θ . **2**
- (d) Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{7} = 1$

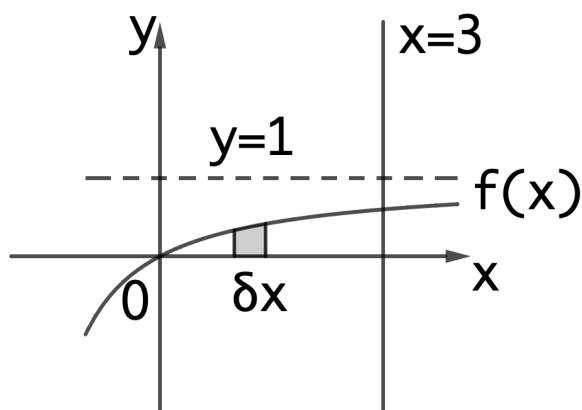


- i. Write down the coordinates of the focus S and the equation of the associated directrix. **2**
- ii. The equation of the normal to the ellipse at the point $P(x_1, y_1)$ is given by **2**

$$\frac{9x}{x_1} - \frac{7y}{y_1} = 2 \quad (\text{DO NOT PROVE THIS.})$$

Let Q be the x -intercept of the normal and let M be the foot of the perpendicular from P to the directrix as shown in the diagram. Show that $QS = \frac{2}{9}PM$.

- (e) The region bounded by the portion of the curve $f(x) = \frac{x}{x+2}$, and the x axis is rotated about the line $x = 3$.



- i. Using the method of cylindrical shells, show that the volume of a typical shell at a distance x from the origin and with distance δx is given by **2**

$$\delta V = 2\pi(3-x)\frac{x}{x+2}\delta x.$$

- ii. Hence, find the volume of this solid. **3**

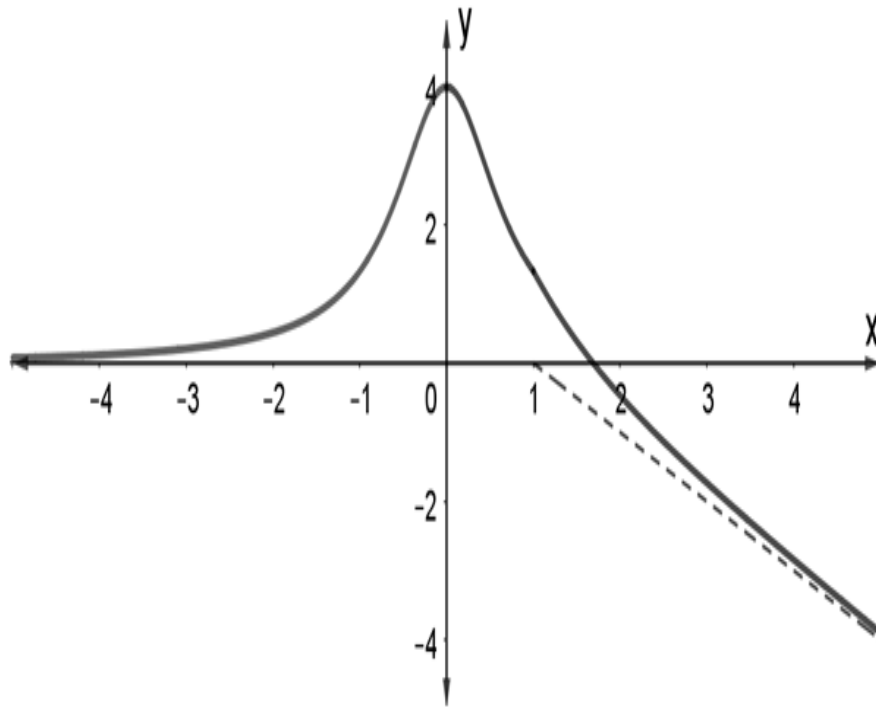
End of Question 12

Question 13 (15 Marks)

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Marks

- (a) The curve $y = f(x)$, sketched below, has asymptotes $y = 0$ and $y = 1 - x$.

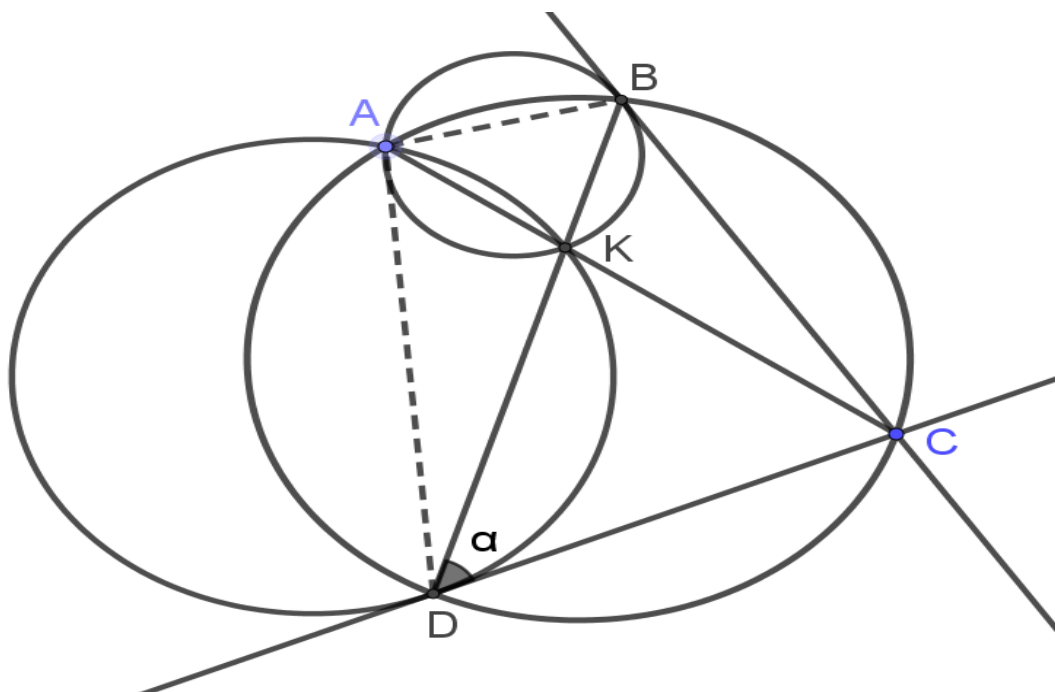


Copy or trace the above graph onto three separate number planes. Use your diagrams to show sketches of the following graphs, showing all essential features clearly.

- i. $y = [f(x)]^2$. **2**
 - ii. $|y| = f(x)$. **2**
 - iii. $y = \log(f(x))$. **2**
- (b) α , β and γ are roots of the cubic equation $x^3 + qx + r = 0$.
- i. Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ in terms of q and r . **2**
 - ii. Find the value of $\alpha^3 + \beta^3 + \gamma^3$ in terms of q and r . **1**
 - iii. By considering $x = (\beta - \gamma)^2$, show that the equation whose roots are $(\beta - \gamma)^2$, $(\gamma - \alpha)^2$ and $(\alpha - \beta)^2$ is **2**

$$(x + q)^3 + 3q(x + q)^2 + 27r^2 = 0.$$

- (c) In the diagram below, $ABCD$ is a cyclic quadrilateral and diagonals AC and BD intersect at K . Circles AKD and AKB are drawn and it is known that CD is a tangent to circle AKD . Let $\angle CDB = \alpha$.



- i. Prove that BCD is isosceles. 2
- ii. Prove that CB is a tangent to circle AKB 2

End of Question 13

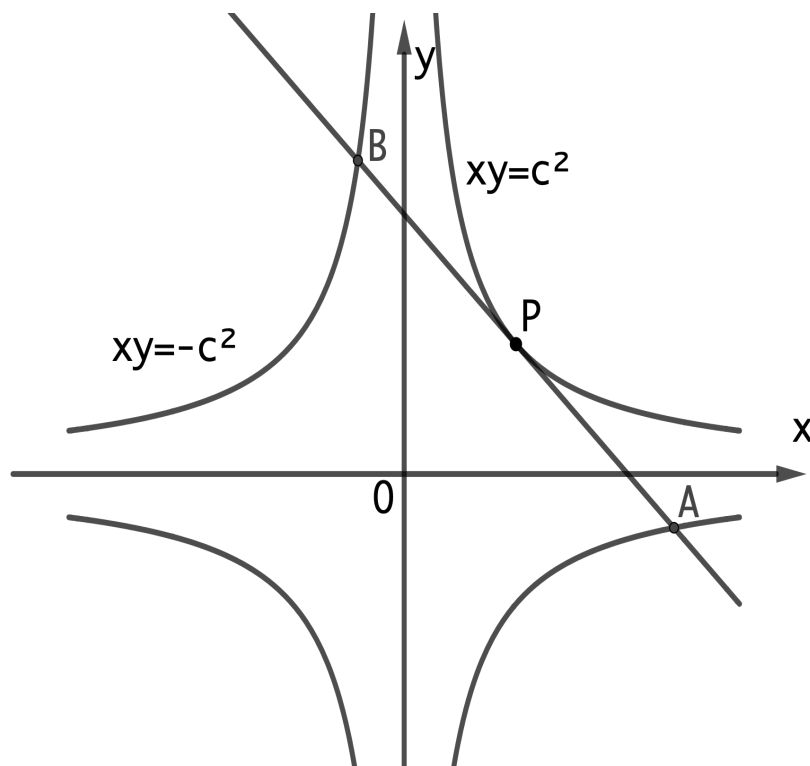
Question 14 (15 Marks)

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Marks

- (a) Consider the rectangular hyperbolas $xy = c^2$ and $xy = -c^2$.
The point $P(cp, c/p)$ lies on $xy = c^2$ and the equation of tangent to $xy = c^2$ at point P is

$$x + p^2y - 2cp = 0.$$



- i. The tangent at P cuts the hyperbola $xy = -c^2$ at two points A and B . Show the coordinates of A and B are $\left(pc(1 + \sqrt{2}), \frac{-c}{p(1 + \sqrt{2})}\right)$ and $\left(pc(1 - \sqrt{2}), \frac{-c}{p(1 - \sqrt{2})}\right)$ respectively. **2**
 - ii. Show that the tangents to $xy = -c^2$ at A and B intersect at $Q(-cp, -c/p)$. **3**
 - iii. Hence, show that the area of the triangle ABQ is independent of p . **3**
- (b) Consider the integral

$$I_n = \int_0^1 \frac{x^n}{\sqrt{x+1}} dx.$$

- i. Show that $I_n + I_{n-1} = \int_0^1 x^{n-1} \sqrt{x+1} dx$ **1**
- ii. Use integration by parts to show that **2**

$$I_n = \frac{-2n}{2n+1} I_{n-1} + \frac{2\sqrt{2}}{2n+1}.$$

- (c) Given that $x^3 + y^3 = 6xy$.
- i. Find the tangent to $x^3 + y^3 = 6xy$ at the point $(3, 3)$. **2**
 - ii. At what point in the first quadrant is the tangent line horizontal? **2**

End of Question 14

Question 15 (15 Marks)

Commence a NEW page.

Marks

- (a) Let α be a real number and suppose z is a complex number such that

$$z + \frac{1}{z} = 2 \cos \alpha.$$

You may assume that

$$z^n + \frac{1}{z^n} = 2 \cos n\alpha$$

for all positive integer n .

- i. Let $\omega = z + \frac{1}{z}$. Prove that

2

$$\omega^3 + \omega^2 - 2\omega - 2 = \left(z + \frac{1}{z}\right) + \left(z^2 + \frac{1}{z^2}\right) + \left(z^3 + \frac{1}{z^3}\right)$$

- ii. Hence, or otherwise, find all solutions of

3

$$\cos \alpha + \cos 2\alpha + \cos 3\alpha = 0, \quad \text{for } 0 \leq \alpha \leq 2\pi.$$

- (b) A sequence x_n is defined by the following rules: $x_0 = 2a$, $x_1 = -a^2$ and

3

$$x_{n+1} = -ax_n + a^2x_{n-1} \quad \text{for } n \geq 1.$$

Prove by mathematical induction that

$$x_n = a^{n+1} \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^n + \left(\frac{-1 - \sqrt{5}}{2} \right)^n \right], \quad \text{for } n \geq 0.$$

- (c) Let a, b and c are positive real numbers.

Given that $x^2 + y^2 \geq 2xy$ for all positive real numbers x and y .

- i. Prove that $a^2 + (bc)^2 \geq 2abc$ **1**
- ii. Prove that $a^2 + b^2 + c^2 \geq ab + bc + ca$ **2**
- iii. Prove that $a^2(1 + b^2) + b^2(1 + c^2) + c^2(1 + a^2) \geq 6abc$ **2**
- iv. Hence, or otherwise prove that $a^2(1 + a^2) + b^2(1 + b^2) + c^2(1 + c^2) \geq 6abc$ **2**

End of Question 15

Question 16 (15 Marks)

Commence a NEW page.

Marks

The recurrence formula is defined by

$$T_0(x) = 2, \quad T_1(x) = 2 \sec x, \quad T_2(x) = 4 \sec^2 x - 2$$

and

$$T_k(x) = 2 \sec x T_{k-1}(x) - T_{k-2}(x) \quad \text{for } k \geq 2 \quad \text{and} \quad 0 \leq x < \frac{\pi}{2}.$$

- i. Show that
- $T_3(x)$
- and
- $T_4(x)$
- are

2

$$T_3(x) = 8 \sec^3 x - 6 \sec x$$

$$T_4(x) = 16 \sec^4 x + 8 \sec^2 x + 2$$

To find a formula for $T_k(x)$, let $F(Z)$ be the power series in Z with the coefficient of Z^k being $T_k(x)$. That is, let

$$F(Z) = 2 + 2 \sec x Z + (4 \sec^2 x - 2) Z^2 + \cdots + T_k(x) Z^k + \cdots$$

- ii. Find
- $(1 - 2 \sec x Z + Z^2)F(Z)$
- , hence show that

3

$$F(Z) = \frac{2 - 2 \sec x Z}{1 - 2 \sec x Z + Z^2}$$

- iii. Given that
- α
- and
- β
- are the zeros of
- $1 - 2 \sec x Z + Z^2 = 0$
- . Show that

1

$$1 - 2 \sec x Z + Z^2 = \left(1 - \frac{Z}{\alpha}\right) \left(1 - \frac{Z}{\beta}\right)$$

- iv. Using partial fraction, show that
- $F(Z)$
- can be written in the form

2

$$F(Z) = \frac{2 - 2 \sec x Z}{1 - 2 \sec x Z + Z^2} = \frac{A}{1 - \frac{Z}{\alpha}} + \frac{B}{1 - \frac{Z}{\beta}}$$

where A and B are constants.

- v. For
- $|Z|$
- sufficiently small, explain why
- $\frac{1}{1 - \frac{Z}{\alpha}}$
- is equal to

2

$$1 + \frac{Z}{\alpha} + \left(\frac{Z}{\alpha}\right)^2 + \cdots + \left(\frac{Z}{\alpha}\right)^k + \cdots,$$

hence show that the coefficient of $T_k(x)$ is $A \left(\frac{1}{\alpha}\right)^k + B \left(\frac{1}{\beta}\right)^k$

- vi. Hence deduce that the formula for
- $T_k(x)$
- is

3

$$T_k(x) = \left(\frac{1}{\sec x + \tan x}\right)^k + \left(\frac{1}{\sec x - \tan x}\right)^k$$

- vii. Find
- $\lim_{n \rightarrow \infty} \frac{T_{n+1}(x)}{T_n(x)}$

2**End of paper.**

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

STUDENT NUMBER:

Class (please ✓)

☐ 12M4A – Mr Ireland

☐ 12M4B – Dr Jomaa

☐ 12M4C – Miss Lee

- | | | | | |
|------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 5 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 6 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 7 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 8 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 9 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 10 – | ☐ A | ☐ B | ☐ C | ☐ D |

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

STUDENT NUMBER:

Class (please ✓)

☐ 12M4A – Mr Ireland

☐ 12M4B – Dr Jomaa

☐ 12M4C – Miss Lee

1 –	☐ A	☒ B	☐ C	☐ D
2 –	☒ A	☐ B	☐ C	☐ D
3 –	☐ A	☐ B	☐ C	☒ D
4 –	☐ A	☐ B	☒ C	☐ D
5 –	☐ A	☒ B	☐ C	☐ D
6 –	☒ A	☐ B	☐ C	☐ D
7 –	☐ A	☐ B	☒ C	☐ D
8 –	☐ A	☐ B	☐ C	☒ D
9 –	☐ A	☐ B	☐ C	☒ D
10 –	☐ A	☐ B	☐ C	☒ D

11. (a) $\int x^4 e^{x^5+3} dx = \frac{1}{5} \int 5x^4 e^{x^5+3} dx = \frac{1}{5} e^{x^5+3} + C$

(b) $|z-2i| = |z|$
 $|z-2i|^2 = |z|^2$
 $x^2 + (y-2)^2 = x^2 + y^2$
 $x^2 + y^2 - 4y + 4 = x^2 + y^2$
 $4 - 4y = 0 \quad \therefore \boxed{y=1}$

(c) $I = \int x \sin^2 x dx, \quad J = \int x \cos^2 x dx$

(i) $I + J = \int x \sin^2 x dx + \int x \cos^2 x dx$
 $= \int x (\sin^2 x + \cos^2 x) dx$
 $= \int x dx$

$\boxed{I + J = \frac{x^2}{2} + C_1}$

(ii) $J - I = \int x \cos^2 x dx - \int x \sin^2 x dx$
 $= \int x (\cos^2 x - \sin^2 x) dx$
 $= \int x \cos 2x dx$

let $u = x$ and $\cos 2x dx = dv$
 $du = dx$ and $v = \frac{1}{2} \sin 2x$

$\int x \cos 2x dx = \frac{1}{2} x \sin 2x - \frac{1}{2} \int \sin 2x dx$
 $= \frac{1}{2} x \sin 2x - \frac{1}{2} \left(-\frac{\cos 2x}{2} \right) + C_2$

$\boxed{J - I = \frac{1}{2} x \sin 2x + \frac{1}{4} \cos 2x + C_2}$

$$(iii) \quad I + J = \frac{x^2}{2} + C_1 \quad (1)$$

$$J - I = \frac{1}{2} x \sin 2x + \frac{1}{4} \cos 2x + C_2 \quad (2)$$

$$(1) + (2) \therefore 2J = \frac{x^2}{2} + \frac{1}{2} x \sin 2x + \frac{1}{4} \cos 2x + C_1 + C_2$$

$$\boxed{J = \frac{x^2}{4} + \frac{1}{4} x \sin 2x + \frac{1}{8} \cos 2x + C}$$

$$(1) - (2) \therefore 2I = \frac{x^2}{2} - \frac{1}{2} x \sin 2x - \frac{1}{4} \cos 2x + C_1 - C_2$$

$$\boxed{I = \frac{x^2}{4} - \frac{1}{4} x \sin 2x - \frac{1}{8} \cos 2x + C}$$



(d) (i) $\int_0^a f(x) dx$

let $x = a - u$

$dx = -du$

$x=0, u=a$

$x=a, u=0$

$$\begin{aligned} \int_0^a f(x) dx &= \int_a^0 f(a-u)(-du) = -\int_a^0 f(a-u) du \\ &= \int_0^a f(a-u) du = \int_0^a f(a-x) dx \end{aligned}$$

(ii) $I = \int_0^{\pi/4} \frac{\cos^n x}{\sin^n x + \cos^n x} dx = \int_0^{\pi/4} \frac{\cos^n(\pi/4 - x)}{\sin^n(\pi/4 - x) + \cos^n(\pi/4 - x)} dx$

$$= \int_0^{\pi/4} \frac{\sin^n(x)}{\cos^n x + \sin^n x} dx$$

$$\begin{aligned} 2I &= \int_0^{\pi/4} \frac{\cos^n x}{\sin^n x + \cos^n x} dx + \int_0^{\pi/4} \frac{\sin^n x}{\sin^n x + \cos^n x} dx \\ &= \int_0^{\pi/4} dx = \left[x \right]_0^{\pi/4} = \pi/4 - 0 \end{aligned}$$

$$I = \pi/4$$

$$\begin{aligned}
\textcircled{e} \quad \int \frac{3x}{5x^2-4x+2} dx &= \frac{3}{10} \int \frac{10x-4+4}{5x^2-4x+2} dx \\
&= \frac{3}{10} \int \frac{10x-4}{5x^2-4x+2} dx + \frac{6}{5} \int \frac{dx}{5x^2-4x+2} \\
&= \frac{3}{10} \ln|5x^2-4x+2| + 6 \int \frac{dx}{(5x)^2-20x+10} \\
&= \frac{3}{10} \ln|5x^2-4x+2| + 6 \int \frac{dx}{(5x-2)^2+6} \\
&= \frac{3}{10} \ln|5x^2-4x+2| + \frac{6}{5} \int \frac{5 dx}{(5x-2)^2+6} \\
&= \frac{3}{10} \ln|5x^2-4x+2| + \frac{6}{5} \times \frac{1}{\sqrt{6}} \tan^{-1} \left(\frac{5x-2}{\sqrt{6}} \right) + C \\
&= \frac{3}{10} \ln|5x^2-4x+2| + \frac{\sqrt{6}}{5} \tan^{-1} \left(\frac{5x-2}{\sqrt{6}} \right) + C
\end{aligned}$$

12.

(a)

$$z = 1 - i$$

$$w^2 = i + 3\bar{z}$$

$$= i + 3(1 + i)$$

$$= 3 + 4i$$

$$= 2^2 - 1^2 + 2 \times 2 \times 1 \times i$$

$$= (2 + i)^2$$

$$w = \pm (2 + i)$$

(b)

$$|z - 3| = 3$$

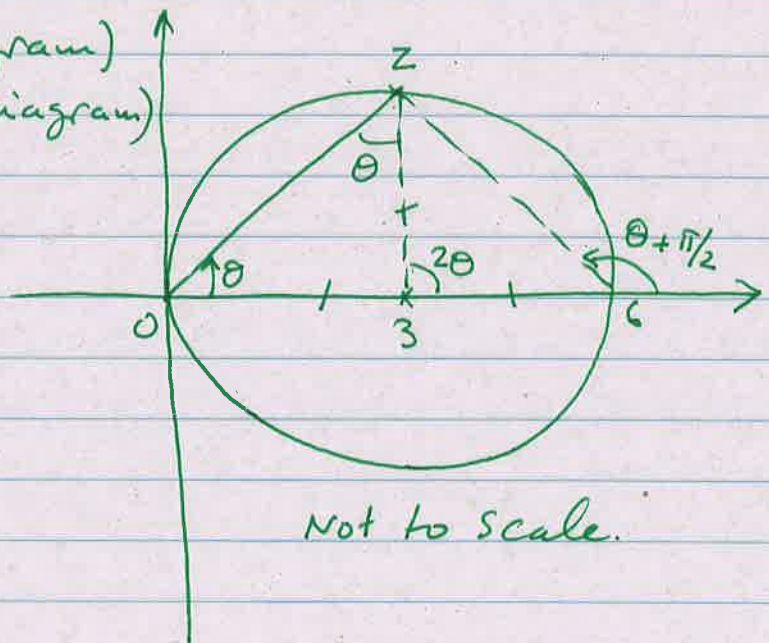
$$\arg(z^2 - 9z + 18) = \arg((z - 3)(z - 6))$$

$$= \arg(z - 3) + \arg(z - 6)$$

$$\arg(z - 3) = 2\theta \quad (\text{see diagram})$$

$$\arg(z - 6) = \theta + \pi/2 \quad (\text{see diagram})$$

$$\arg(z^2 - 9z + 18) = \pi/2 + 3\theta$$



Not to scale.

(c)

$$\frac{x^2}{9} + \frac{y^2}{7} = 1$$

$$(i) \quad 7 = 9(1 - e^2) \therefore \frac{7}{9} = 1 - e^2 \therefore e^2 = 1 - \frac{7}{9} = \frac{2}{9} \therefore e = \frac{\sqrt{2}}{3}$$

$$a = 3$$

$$s(ae, 0) = (\sqrt{2}, 0)$$

$$\text{the directrix } x = + \frac{a}{e} = + \frac{3}{\frac{\sqrt{2}}{3}} = + \frac{9}{\sqrt{2}}$$

$$(ii) \quad \frac{qx}{x_1} - \frac{7y}{y_1} = 2$$

$$\text{Sub } y=0 \therefore x = \frac{2x_1}{q} \quad \therefore Q\left(\frac{2x_1}{q}, 0\right)$$

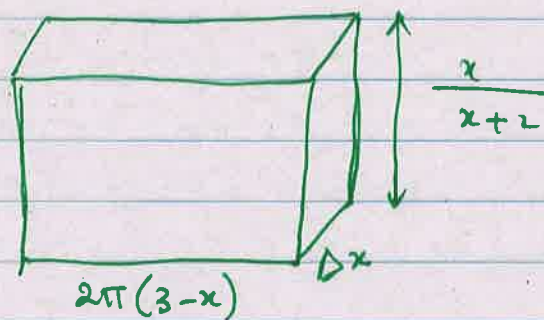
$$\text{Also, } M\left(\frac{q}{\sqrt{2}}, y\right)$$

$$QS = \sqrt{2} - \frac{2x_1}{q}$$

$$PM = \frac{q}{\sqrt{2}} - x_1$$

$$QS = \sqrt{2} - \frac{2x_1}{q} = \frac{2}{q} \left(\frac{q}{\sqrt{2}} - x_1 \right) = \frac{2}{q} PM.$$

(d)



$$\text{Area} = 2\pi(3-x) \frac{x}{x+2}$$

$$\delta V = 2\pi(3-x) \frac{x}{x+2} \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \delta V = \lim_{\Delta x \rightarrow 0} 2\pi(3-x) \frac{x}{x+2} \Delta x$$

$$V = 2\pi \int_0^3 (3-x) \frac{x}{x+2} dx = 2\pi \int_0^3 (3-x) \left(\frac{x+2-2}{x+2} \right) dx$$

$$= 2\pi \int_0^3 (3-x) dx + 4\pi \int_0^3 \frac{x-3}{x+2} dx$$

$$= 2\pi \int_0^3 (3-x) dx + 4\pi \int_0^3 dx - 20\pi \int_0^3 \frac{dx}{x+2}$$

$$= 2\pi \left[3x - \frac{x^2}{2} \right]_0^3 + 4\pi (x)_0^3 - 20\pi \left[\ln|x+2| \right]_0^3$$

continue (d)

$$V = 2\pi \left(9 - \frac{3^2}{2} - 0 \right) + 4\pi (3 - 0) - 20\pi (\ln 5 - \ln 2)$$

$$= 9\pi + 12\pi - 20\pi \ln \left(\frac{5}{2} \right)$$

$$= 21\pi - 20\pi \ln \left(\frac{5}{2} \right)$$

$$\boxed{V = \pi \left(21 - 20 \ln \frac{5}{2} \right)}$$

(e) $\int \frac{dx}{x(1+x^2)}$

let $u = \frac{1}{1+x^2} \therefore du = \frac{-2x}{(1+x^2)^2} dx$

$$1+x^2 = \frac{1}{u} \therefore x^2 = \frac{1}{u} - 1 = \frac{1-u}{u} \therefore \frac{1}{x^2} = \frac{u}{1-u}$$

$$\int \frac{dx}{x(1+x^2)} = -\frac{1}{2} \int \frac{-2x dx}{x^2(1+x^2)^2} = -\frac{1}{2} \int \frac{u du}{1-u}$$

$$= \frac{1}{2} \int \frac{1-u-1}{1-u} du = \frac{1}{2} \int du - \frac{1}{2} \int \frac{du}{1-u}$$

$$= \frac{1}{2} u + \frac{1}{2} \ln |1-u| + C$$

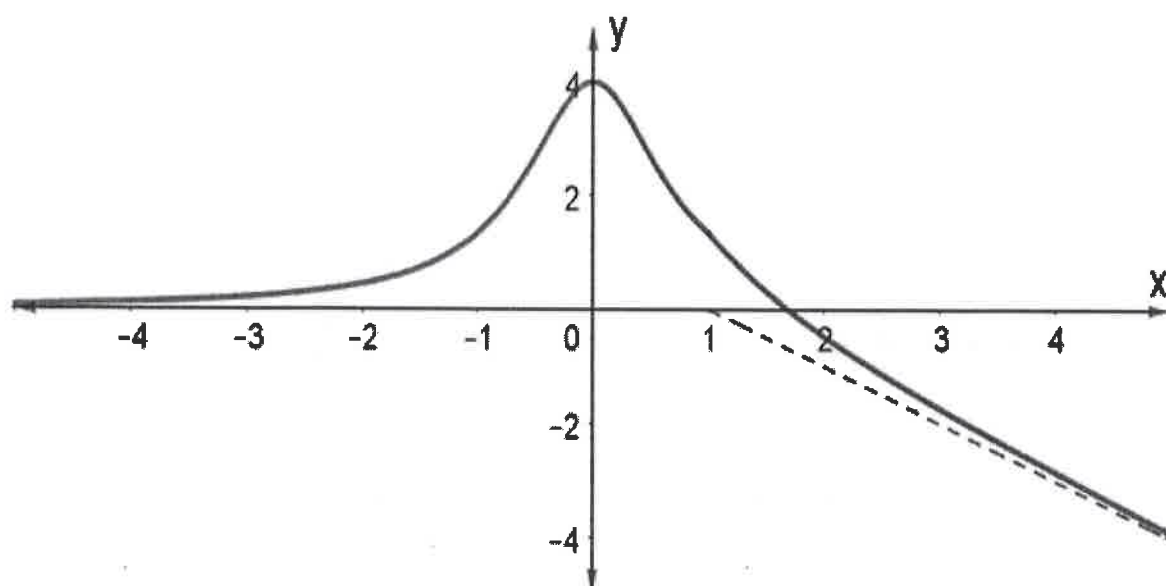
$$= \frac{1}{2} \times \frac{1}{1+x^2} + \frac{1}{2} \ln \left| 1 - \frac{1}{1+x^2} \right| + C$$

$$= \frac{1}{2(1+x^2)} + \frac{1}{2} \ln \left(\frac{x^2}{1+x^2} \right) + C$$

Question 13 :

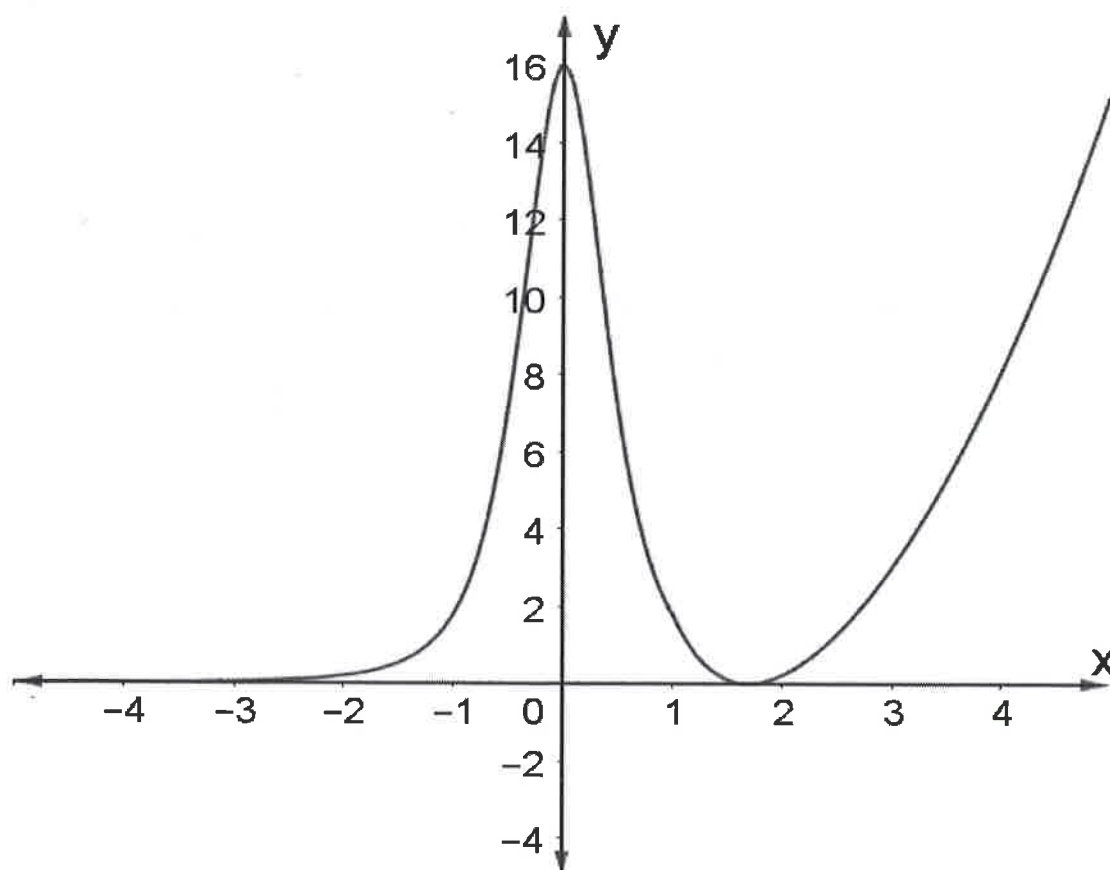
(a)

$$Y=f(x)$$

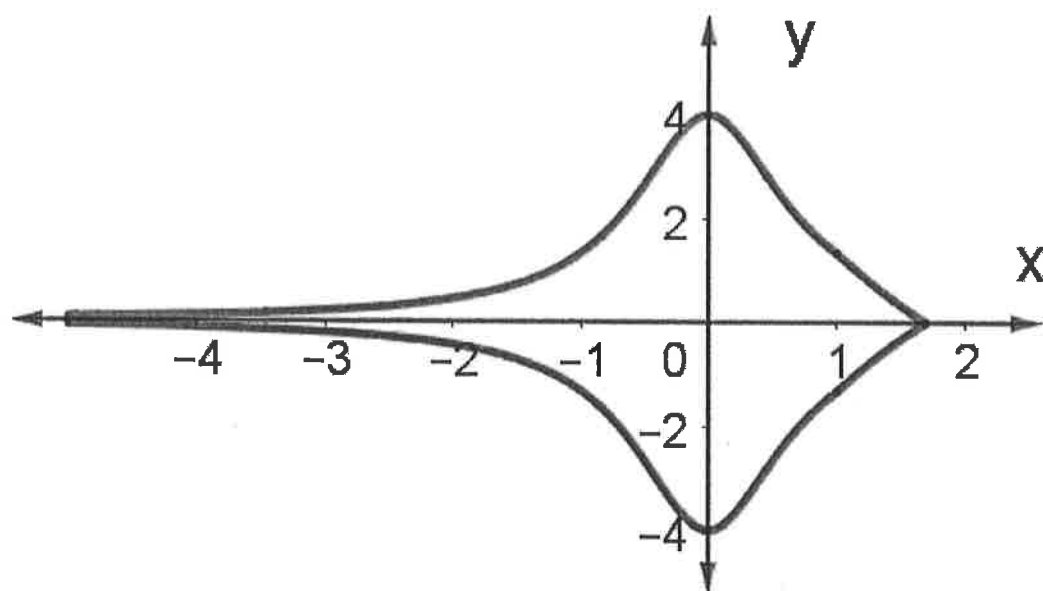


(i)

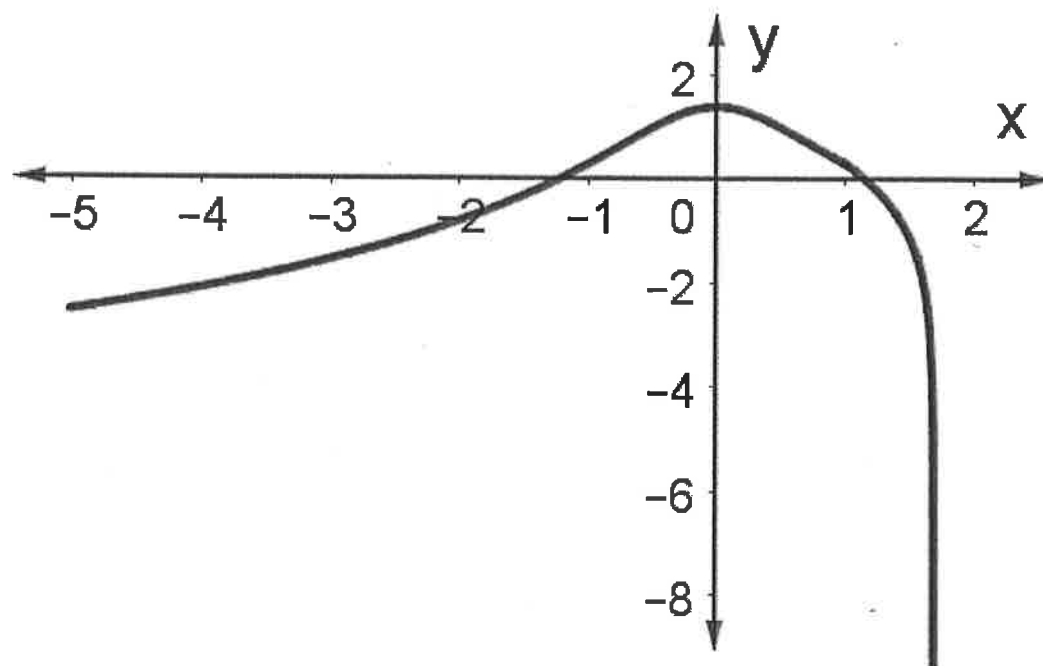
$$Y=f(x)^2$$



(ii) $|y|=f(x)$



(iii) $Y=\log(f(x))$



(b)

$$x^3 + qx + r = 0$$

α, β and γ are roots

$$\therefore \alpha + \beta + \gamma = 0 \quad (1)$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = q \quad (2)$$

$$\alpha\beta\gamma = -r \quad (3)$$

$$(i) \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{q}{-r} = -\frac{q}{r}$$

$$(ii) \quad \alpha^3 + \beta^3 + \gamma^3 = -q(\alpha + \beta + \gamma) - 3r \\ = -q \times 0 - 3r \\ = -3r$$

$$(iii) \quad x = (\beta - \gamma)^2 = (\beta + \gamma)^2 - 4\beta\gamma$$

$$(1) \therefore \beta + \gamma = -\alpha$$

$$(3) \therefore \beta\gamma = -r/\alpha$$

$$x = (-\alpha)^2 - 4(-r/\alpha) = \alpha^2 + \frac{4r}{\alpha} \therefore \alpha^3 - \alpha x + 4r = 0 \quad (4)$$

$$\text{but, } \alpha^3 + q\alpha + r = 0 \quad (5)$$

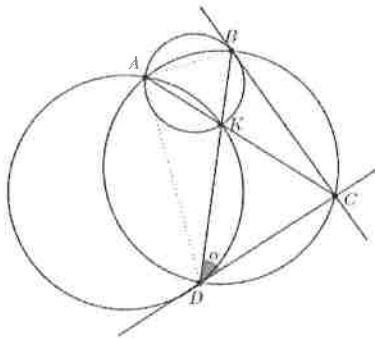
$$(5) - (4) \therefore (q+x)\alpha = 3r \therefore \alpha = \frac{3r}{q+x} \quad (6)$$

sub (6) into (5) we obtain:

$$\frac{27r^3}{(q+x)^3} + \frac{3qr}{q+x} + r = 0$$

$$27r^3 + 3q(q+x)^2 + (q+x)^3 = 0$$

$$\text{OR} \quad x^3 + 6qx^2 + 9q^2x + 4q^3 + 27r^2 = 0.$$



In the diagram above, $ABCD$ is a cyclic quadrilateral and diagonals AC and BD intersect at K . Circles AKD and AKB are drawn and it is known that CD is a tangent to circle AKD . Let $\angle CDB = \alpha$.

Use the separate blue answer sheet for Question 16 (b).

- | | | |
|------|--|---|
| (i) | Prove that $\triangle BCD$ is isosceles. | 2 |
| (ii) | Prove that CB is a tangent to circle AKB . | 2 |

(c) (i) If $\angle CDB = \alpha$
 then $\angle DAC = \alpha$ (Angle in the alternate segment theorem)
 and $\therefore \angle DBC = \alpha$ (Angles on the circumference standing on the same arc are equal)
 $\therefore \triangle BCD$ is isosceles (Base $\angle$'s equal).

(ii) $CD^2 = AC \cdot CK$ (Square on tangent).
 but $CD = BC$ (Equal sides of isosceles $\triangle BCD$)
 $\therefore BC^2 = AC \cdot CK$
 $\therefore BC$ must be a tangent to circle AKB .

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(a)

$$x^3 + y^3 = 6xy$$

$$(i) \quad 3x^2 + 3y^2 \frac{dy}{dx} = 6y + 6x \frac{dy}{dx}$$

$$x^2 + y^2 \frac{dy}{dx} = 2y + 2x \frac{dy}{dx}$$

$$(y^2 - 2x) \frac{dy}{dx} = 2y - x^2$$

$$\frac{dy}{dx} = \frac{2y - x^2}{y^2 - 2x}$$

$$\text{at } (3, 3), \quad \frac{dy}{dx} = \frac{2(3) - 3^2}{3^2 - 2(3)} = \frac{-3}{3} = -1$$

The equation of tangent is

$$y - 3 = -1(x - 3) \therefore \boxed{y = -x + 6}$$

(ii) Tangent line is horizontal $\therefore y' = 0$

$$\therefore 2y - x^2 = 0 \therefore y = x^2/2$$

$$x^3 + \left(\frac{x^2}{2}\right)^3 = 6x\left(\frac{x^2}{2}\right)$$

$$x^3 + \frac{x^6}{8} = 3x^3$$

$$x^6 = 16x^3$$

$$x^3 = 16$$

$$x = 16^{1/3} = 2^{4/3}$$

$$y = \frac{(2^{4/3})^2}{2} = 2^{5/3}$$

$\therefore$ At $(2^{4/3}, 2^{5/3})$ the tangent line is horizontal.

(b) (i) $x + p^2y - 2cp = 0$
multiply by x , we obtain

$$x^2 + p^2xy - 2cp x = 0, \text{ but } xy = -c^2$$

$$x^2 - c^2p^2 - 2cp x = 0$$

$$x^2 - 2cp x + c^2p^2 - 2c^2p^2 = 0$$

$$(x - cp)^2 = 2c^2p^2$$

$$x - cp = \pm cp\sqrt{2}$$

$$x = cp(1 \pm \sqrt{2})$$

$$\therefore A \left(cp(1 + \sqrt{2}), \frac{-c}{p(1 + \sqrt{2})} \right)$$

$$\text{and } B \left(cp(1 - \sqrt{2}), \frac{-c}{p(1 - \sqrt{2})} \right)$$

(ii) The equation of tangent at A is

$$y + \frac{c}{p(1 + \sqrt{2})} = \frac{c^2}{c^2p^2(1 + \sqrt{2})^2} (x - cp(1 + \sqrt{2}))$$

$$\boxed{y = \frac{1}{p^2(3 + 2\sqrt{2})}x - \frac{2c}{p(1 + \sqrt{2})}} \quad (1)$$

Similarly at B is

$$\boxed{y = \frac{1}{p^2(3 - 2\sqrt{2})}x - \frac{2c}{p(1 - \sqrt{2})}} \quad (2)$$

$$\begin{aligned} (1) - (2) \quad \therefore 0 &= \frac{x}{p^2} \left(\frac{1}{3 + 2\sqrt{2}} - \frac{1}{3 - 2\sqrt{2}} \right) - \frac{2c}{p} \left(\frac{1}{1 + \sqrt{2}} - \frac{1}{1 - \sqrt{2}} \right) \\ &= \frac{x}{p^2} \left(\frac{-4\sqrt{2}}{1} \right) - \frac{2c}{p} \left(\frac{-2\sqrt{2}}{-1} \right) \\ &= \frac{-4\sqrt{2}}{p^2}x - \frac{4\sqrt{2}c}{p} \quad \therefore \boxed{x = -cp} \end{aligned}$$

(b) continue
use (i)

$$y = \frac{1}{p^2(3+2\sqrt{2})}(-cp) - \frac{2c}{p(1+\sqrt{2})}$$

$$= \frac{-c}{p} \times \frac{3-2\sqrt{2}}{9-8} - \frac{2c}{p} \frac{(1-\sqrt{2})}{(1-2)}$$

$$= \frac{-c}{p} (3-2\sqrt{2} - 2+2\sqrt{2}) = \frac{-c}{p}$$

$\therefore Q(-cp, -c/p)$ as required.

$$\begin{aligned} \text{(iii)} \quad AB &= \sqrt{[cp(1+\sqrt{2}) - cp(1-\sqrt{2})]^2 + \left[\frac{-c}{p(1+\sqrt{2})} - \frac{-c}{p(1-\sqrt{2})}\right]^2} \\ &= \sqrt{8p^2c^2 + \frac{8c^2}{p^2}} \\ &= 2\sqrt{2}c \sqrt{p^2 + \frac{1}{p^2}} \end{aligned}$$

Equation of AB is $x^2 + p^2y - 2cp = 0$

perpendicular distance from Q to AB is

$$\frac{|-cp + p^2(-c/p) - 2cp|}{\sqrt{1+p^4}} = \frac{4cp}{\sqrt{p^4+1}}$$

$$\begin{aligned} \text{Area of } ABQ &= \frac{1}{2} \times \frac{4cp}{\sqrt{p^4+1}} \times 2\sqrt{2}c \sqrt{p^2 + \frac{1}{p^2}} \\ &= 4c^2\sqrt{2} \end{aligned}$$

is independent of p as required.

$$(6) (i) I_n = \int_0^1 \frac{x^n}{\sqrt{x+1}} dx$$

$$I_{n-1} = \int_0^1 \frac{x^{n-1}}{\sqrt{x+1}} dx$$

$$\begin{aligned} I_n + I_{n-1} &= \int_0^1 \frac{x^{n-1}(1+x)}{\sqrt{x+1}} dx \\ &= \int_0^1 x^{n-1} \sqrt{1+x} dx \end{aligned}$$

$$(ii) \text{ let } x^{n-1} dx = dv \therefore v = \frac{1}{n} x^n$$

$$u = \sqrt{1+x} \therefore du = \frac{1}{2\sqrt{1+x}} dx$$

$$\begin{aligned} \int_0^1 x^{n-1} \sqrt{1+x} dx &= \left[\frac{1}{n} x^n \sqrt{1+x} \right]_0^1 - \frac{1}{2n} \int_0^1 \frac{x^n}{\sqrt{1+x}} dx \\ &= \frac{1}{n} \sqrt{2} - \frac{1}{2n} I_n \end{aligned}$$

$$I_n + I_{n-1} = \frac{1}{n} \sqrt{2} - \frac{1}{2n} I_n$$

$$2n I_n + 2n I_{n-1} = 2\sqrt{2} - I_n$$

$$(2n+1) I_n = -2n I_{n-1} + 2\sqrt{2}$$

$$I_n = \frac{-2n}{2n+1} I_{n-1} + \frac{2\sqrt{2}}{2n+1}$$

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$$\begin{aligned}
 (i) \quad & \omega^3 + \omega^2 - 2\omega - 2 \\
 &= \omega^2(\omega + 1) - 2(\omega + 1) \\
 &= (\omega + 1)(\omega^2 - 2) \\
 &= \left(z + \frac{1}{z} + 1\right) \left(\left(z + \frac{1}{z}\right)^2 - 2\right) \\
 &= \left(z + \frac{1}{z} + 1\right) \left(z^2 + \frac{1}{z^2}\right) \\
 &= z^3 + \frac{1}{z} + z + \frac{1}{z^3} + z^2 + \frac{1}{z^2} \\
 &= z + \frac{1}{z} + z^2 + \frac{1}{z^2} + z^3 + \frac{1}{z^3}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad & z + \frac{1}{z} = 2\cos\alpha \\
 & z^2 + \frac{1}{z^2} = 2\cos 2\alpha \\
 & z^3 + \frac{1}{z^3} = 2\cos 3\alpha
 \end{aligned}$$

$$\begin{aligned}
 \omega^3 + \omega^2 - 2\omega - 2 &= (\omega + 1)(\omega^2 - 2) \\
 &= (2\cos\alpha + 1)(2\cos 2\alpha) \\
 &= 2\cos 2\alpha(2\cos\alpha + 1)
 \end{aligned}$$

$$\cos\alpha + \cos 2\alpha + \cos 3\alpha = 0 \quad \therefore \quad \frac{1}{2} \left[z + \frac{1}{z} + z^2 + \frac{1}{z^2} + z^3 + \frac{1}{z^3} \right] = 0$$

$$\therefore \frac{1}{2}(\omega + 1)(\omega^2 - 2) = 0$$

$$\therefore \cos 2\alpha(2\cos\alpha + 1) = 0$$

$$\cos 2\alpha = 0 \quad \text{or} \quad 2\cos\alpha + 1 = 0$$

$$2\alpha = \frac{\pi}{2} \quad \text{or} \quad \frac{3\pi}{2} \quad \text{or} \quad \cos\alpha = -\frac{1}{2} \quad \therefore \alpha = \frac{2\pi}{3} \quad \text{or} \quad \frac{4\pi}{3}$$

$$\frac{5\pi}{2} \quad \text{or} \quad \frac{7\pi}{2}$$

$$\alpha = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$(b) \quad x_0 = a^{0+1} \left[\left(\frac{-1+\sqrt{5}}{2} \right)^0 + \left(\frac{-1-\sqrt{5}}{2} \right)^0 \right] = a(1+1) = 2a$$

$$x_1 = a^2 \left[\frac{-1+\sqrt{5}}{2} + \frac{-1-\sqrt{5}}{2} \right] = a^2 \left(-\frac{1}{2} - \frac{1}{2} \right) = -a^2.$$

Assume it is true for $n=k-1$ and $n=k$

$$\left. \begin{aligned} x_{k-1} &= a^k \left[\left(\frac{-1+\sqrt{5}}{2} \right)^{k-1} + \left(\frac{-1-\sqrt{5}}{2} \right)^{k-1} \right] \\ x_k &= a^{k+1} \left[\left(\frac{-1+\sqrt{5}}{2} \right)^k + \left(\frac{-1-\sqrt{5}}{2} \right)^k \right] \end{aligned} \right\} (+) \text{ (induction hypothesis)}$$

prove it true for $n=k+1$.

$$x_{k+1} = -a x_k + a^2 x_{k-1}$$

$$= -a a^{k+1} \left[\left(\frac{-1+\sqrt{5}}{2} \right)^k + \left(\frac{-1-\sqrt{5}}{2} \right)^k \right] + a^2 a^k \left[\left(\frac{-1+\sqrt{5}}{2} \right)^{k-1} + \left(\frac{-1-\sqrt{5}}{2} \right)^{k-1} \right]$$

$$= a^{k+2} \left[\left(\frac{-1+\sqrt{5}}{2} \right)^{k-1} \left(1 - \frac{-1+\sqrt{5}}{2} \right) + \left(\frac{-1-\sqrt{5}}{2} \right)^{k-1} \left(1 - \frac{-1-\sqrt{5}}{2} \right) \right]$$

$$= a^{k+2} \left[\left(\frac{-1+\sqrt{5}}{2} \right)^{k-1} \left(\frac{3-\sqrt{5}}{2} \right) + \left(\frac{-1-\sqrt{5}}{2} \right)^{k-1} \left(\frac{3+\sqrt{5}}{2} \right) \right]$$

$$\boxed{\begin{aligned} \left(\frac{-1+\sqrt{5}}{2} \right)^2 &= \frac{1+5-2\sqrt{5}}{4} = \frac{6-2\sqrt{5}}{4} = \frac{3-\sqrt{5}}{2} \\ \left(\frac{-1-\sqrt{5}}{2} \right)^2 &= \frac{1+5+2\sqrt{5}}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2} \end{aligned}}$$

$$\begin{aligned} x_{k+1} &= a^{k+2} \left[\left(\frac{-1+\sqrt{5}}{2} \right)^{k-1} \left(\frac{-1+\sqrt{5}}{2} \right)^2 + \left(\frac{-1-\sqrt{5}}{2} \right)^{k-1} \left(\frac{-1-\sqrt{5}}{2} \right)^2 \right] \\ &= a^{k+2} \left[\left(\frac{-1+\sqrt{5}}{2} \right)^{k+1} + \left(\frac{-1-\sqrt{5}}{2} \right)^{k+1} \right] \end{aligned}$$

Hence by mathematical induction, it is true for all $n \geq 0$.

$$(c) \quad x^2 + y^2 \geq 2xy$$

$$(i) \quad a^2 + (bc)^2 \geq 2a(bc) = 2abc.$$

$$(ii) \quad \left. \begin{array}{l} a^2 + b^2 \geq 2ab \\ a^2 + c^2 \geq 2ac \\ b^2 + c^2 \geq 2bc \end{array} \right\} a^2 + b^2 + c^2 \geq ab + bc + ca.$$

$$(iii) \quad \begin{aligned} & a^2(1+b^2) + b^2(1+c^2) + c^2(1+a^2) \\ &= a^2 + a^2b^2 + b^2 + b^2c^2 + c^2 + c^2a^2 \\ &= a^2 + b^2 + c^2 + a^2b^2 + b^2c^2 + c^2a^2 \end{aligned}$$

$$\begin{aligned} & a^2 + b^2 + c^2 \geq 3abc \\ & a^2b^2 + b^2c^2 + c^2a^2 \geq 3abc \end{aligned} \quad \geq 6abc.$$

$$(iv) \quad \begin{aligned} & a^4 + b^4 + c^4 + a^2 + b^2 + c^2 \\ & \geq a^2b^2 + b^2c^2 + c^2a^2 + a^2 + b^2 + c^2 \\ & \geq 6abc. \end{aligned}$$

Question 16:

$$T_0(x) = 2, \quad T_1(x) = 2 \sec x, \quad T_2(x) = 4 \sec^2 x - 2$$

$$T_k(x) = 2 \sec x T_{k-1}(x) - T_{k-2}(x) \quad k \geq 2, \quad 0 \leq x < \pi/2 \quad (*)$$

$$\begin{aligned} \text{(i)} \quad T_3(x) &= 2 \sec x T_2(x) - T_1(x) \\ &= 2 \sec x (4 \sec^2 x - 2) - 2 \sec x \\ &= 8 \sec^3 x - 4 \sec x - 2 \sec x \\ &= 8 \sec^3 x - 6 \sec x \end{aligned}$$

$$\begin{aligned} T_4(x) &= 2 \sec x T_3(x) - T_2(x) \\ &= 2 \sec x (8 \sec^3 x - 6 \sec x) - (4 \sec^2 x - 2) \\ &= 16 \sec^4 x - 12 \sec^2 x - 4 \sec^2 x + 2 \\ &= 16 \sec^4 x - 16 \sec^2 x + 2 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad F(z) &= 2 + 2 \sec x z + (4 \sec^2 x - 2) z^2 + \dots + T_k(x) z^k + \dots \\ -2 \sec x z F(z) &= -4 \sec x z - 4 \sec^3 x z^2 - 2 \sec x z^3 T_2(x) + \dots - 2 \sec x T_{k-1}(x) z^k + \dots \\ z^2 F(z) &= 2 z^2 + T_1(x) z^3 + \dots + T_{k-2}(x) z^k + \dots \end{aligned}$$

$$\begin{aligned} (1 - 2 \sec x z + z^2) F(z) &= 2 + 2 \sec x z + 0 z^2 + (T_3(x) - 2 \sec x T_2(x) + T_1(x)) z^3 + \dots \\ &\quad + (T_k(x) - 2 \sec x T_{k-1}(x) + T_{k-2}(x)) z^k + \dots \\ &= 2 - 2 \sec x z \end{aligned}$$

Coefficients of z^k , $k \geq 3$ are zero because they satisfy the recurrence formula (*)

$$\therefore F(z) = \frac{2 - 2 \sec x z}{1 - 2 \sec x z + z^2}$$

$$\begin{aligned} \text{(ii)} \quad \alpha, \beta \text{ are the roots of } 1 - 2 \sec x z + z^2 &= 0 \quad \therefore \alpha \beta = 1 \\ 1 - 2 \sec x z + z^2 &= (z - \alpha)(z - \beta) = \frac{(z - \alpha)(z - \beta)}{\alpha \beta} \\ &= \left(\frac{z}{\alpha} - 1\right) \left(\frac{z}{\beta} - 1\right) \\ &= \left(1 - \frac{z}{\alpha}\right) \left(1 - \frac{z}{\beta}\right) \end{aligned}$$

using partial fraction.

$$(i) \quad F(z) = \frac{2 - 2\sec x z}{1 - 2\sec x z + z^2} = \frac{2 - 2\sec x z}{(1 - \frac{z}{\alpha})(1 - \frac{z}{\beta})} = \frac{A}{1 - \frac{z}{\alpha}} + \frac{B}{1 - \frac{z}{\beta}}$$

$$= \frac{A(1 - \frac{z}{\beta}) + B(1 - \frac{z}{\alpha})}{(1 - \frac{z}{\alpha})(1 - \frac{z}{\beta})} = \frac{A + B - (\frac{A}{\beta} + \frac{B}{\alpha})z}{(1 - \frac{z}{\alpha})(1 - \frac{z}{\beta})}$$

$$\therefore A + B = 2 \quad \therefore B = 2 - A \quad (1)$$

$$\frac{A}{\beta} + \frac{B}{\alpha} = 2\sec x \quad \therefore \alpha A + \beta B = 2\sec x \quad (2) \quad (\alpha\beta = 1)$$

$$\begin{pmatrix} 1 & 1 \\ \alpha & \beta \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 2 \\ 2\sec x \end{pmatrix}$$

$$\alpha + \beta = 2\sec x \quad (\text{sum of roots})$$

$$\therefore A = B = 1.$$

(v). $|z| \ll 1$, $\frac{1}{1 - \frac{z}{\alpha}}$ is the limiting sum of the infinite geometric series

$$1 + \frac{z}{\alpha} + \left(\frac{z}{\alpha}\right)^2 + \dots + \left(\frac{z}{\alpha}\right)^k + \dots$$

$$F(z) = \frac{A}{1 - \frac{z}{\alpha}} + \frac{B}{1 - \frac{z}{\beta}} = A \left(1 + \frac{z}{\alpha} + \left(\frac{z}{\alpha}\right)^2 + \dots + \left(\frac{z}{\alpha}\right)^k + \dots \right) + B \left(1 + \frac{z}{\beta} + \left(\frac{z}{\beta}\right)^2 + \dots + \left(\frac{z}{\beta}\right)^k + \dots \right)$$

But $A = B = 1$

$$= 2 + \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)z + \left(\frac{1}{\alpha^2} + \frac{1}{\beta^2}\right)z^2 + \dots + \left(\left(\frac{1}{\alpha}\right)^k + \left(\frac{1}{\beta}\right)^k\right)z^k + \dots$$

$$\therefore T_k(x) = \left(\frac{1}{\alpha}\right)^k + \left(\frac{1}{\beta}\right)^k$$

(vi) From the quadratic equation $1 - 2\sec x z + z^2 = 0$

$$\alpha = \frac{2\sec x + \sqrt{4\sec^2 x - 4}}{2} = \sec x + \tan x$$

$$\beta = \frac{2\sec x - \sqrt{4\sec^2 x - 4}}{2} = \sec x - \tan x$$

$$T_k(n) = \left(\frac{1}{\sec n + \tan n} \right)^k + \left(\frac{1}{\sec n - \tan n} \right)^k$$

$$= (\sec n - \tan n)^k + (\sec n + \tan n)^k$$

$$(Vii) \lim_{n \rightarrow \infty} \frac{T_{n+1}(n)}{T_n(n)}$$

$$= \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{\sec n + \tan n} \right)^{n+1} + \left(\frac{1}{\sec n - \tan n} \right)^{n+1}}{\left(\frac{1}{\sec n + \tan n} \right)^n + \left(\frac{1}{\sec n - \tan n} \right)^n}$$

$$= \lim_{n \rightarrow \infty} \frac{(\sec n + \tan n)^{n+1} + (\sec n - \tan n)^{n+1}}{(\sec n + \tan n)^n + (\sec n - \tan n)^n}$$

$$= \lim_{n \rightarrow \infty} \frac{(\sec n + \tan n)^{n+1} \left[1 + \left(\frac{\sec n - \tan n}{\sec n + \tan n} \right)^{n+1} \right]}{(\sec n + \tan n)^n \left[1 + \left(\frac{\sec n - \tan n}{\sec n + \tan n} \right)^n \right]}$$

$$\boxed{\frac{\sec n - \tan n}{\sec n + \tan n} < 1, \text{ As } n \rightarrow \infty, \left(\frac{\sec n - \tan n}{\sec n + \tan n} \right)^n \rightarrow 0}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{T_{n+1}(n)}{T_n(n)} = \sec n + \tan n.$$
